

SMART PEOPLE COUNTER SYSTEM WITH AUTOMATED LIGHT CONTROL

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- **Department:** BS-AI
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Introduction and project overview:

The DLD Smart People Counter System with Automated Light Control is an intelligent embedded system designed to count the number of people entering and exiting an indoor space and automatically control lighting based on occupancy.

The system uses infrared (IR) sensors to detect movement, an Arduino Uno microcontroller for processing, Bluetooth communication for wireless data transfer, and an Android application for Real-time monitoring.

The main goal of this project is to provide an accurate, low-cost, and energy-efficient solution for occupancy monitoring in places such as offices, classrooms, libraries, and retail stores. Unlike manual counting or expensive commercial systems, this project integrates people counting and lighting control into a single smart solution.

Hardware components and specifications:

The system consists of the following hardware components:

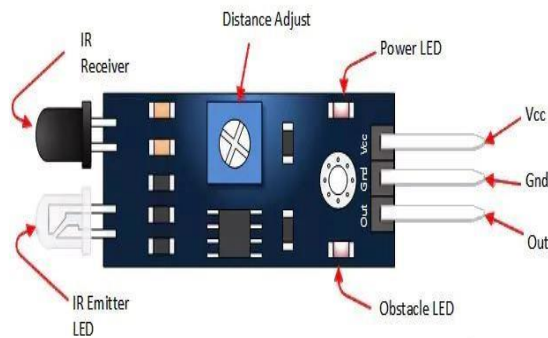
Arduino Uno

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Clock Speed: 16 MHz
- Function: Main controller for processing sensor data and controlling outputs

IR Sensor Modules (TCRT5000)

- Quantity: 2
- Operating Voltage: 3.3V–5V
- Function: Detect entry and exit of people

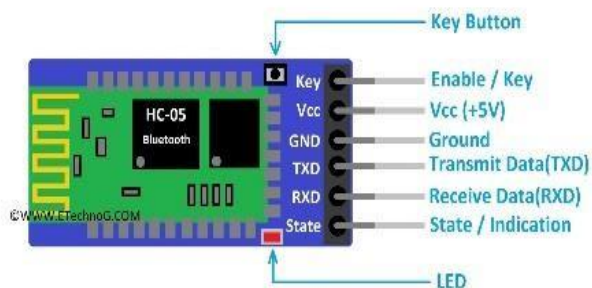
- Feature: Adjustable sensitivity using onboard potentiometer
-



HC-05 Bluetooth Module

- Operating Voltage: 3.3V–5V
- Range: Up to 10 meters
- Function: Wireless communication between Arduino and Android app

Figure.01: Bluetooth Module HC-05 Pinout Diagram



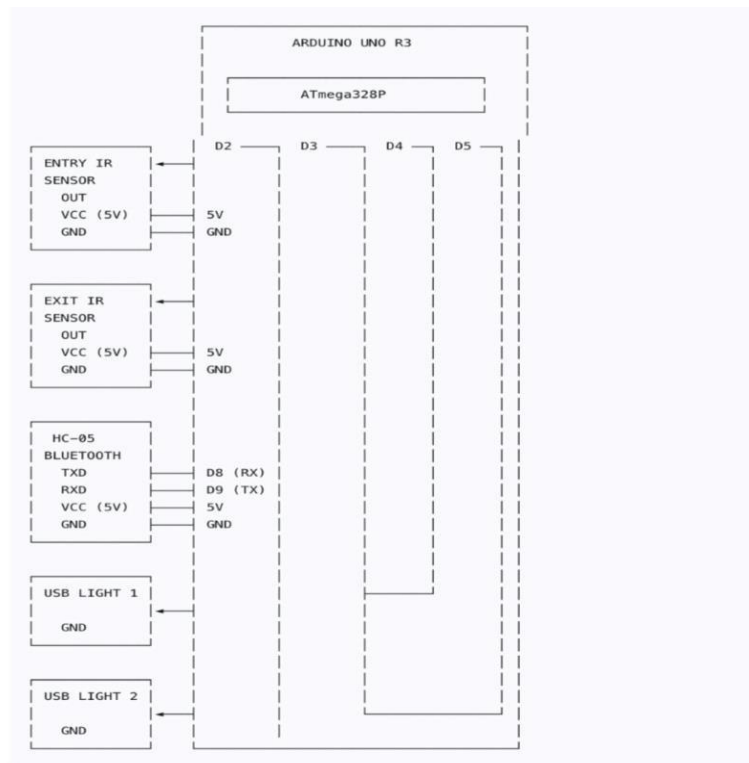
USB LED Lights

- Voltage: 5V
- Function: Lighting control based on occupancy

Other Components

- Breadboard, jumper wires, resistors (220Ω, 10kΩ)
- USB cable and power supply (5V)

Circuit diagram:



Software:

The software part of the project includes:

Arduino Software

- Language: Embedded C/C++
- Platform: Arduino IDE
- Responsibilities:
 - Reading sensor inputs
 - Detecting direction (entry/exit)
 - Updating people count
 - Controlling lights
 - Sending data via Bluetooth

Android Application

- Displays real-time people count
- Shows light status
- Allows manual light ON/OFF
- Sends control commands to Arduino

The system uses a **state-machine based logic** to ensure reliable detection and prevent false triggers.

Working principle and algorithm:

Working Principle

- IR sensors detect reflected infrared light when a person crosses the doorway.
- Arduino determines the **sequence of sensor activation** to identify entry or exit.
- People count is updated accordingly.
- Lights turn **ON when count > 0** and **OFF when count = 0**.
- Data is transmitted to the Android app via Bluetooth.

Algorithm (Simplified)

1. Initialize sensors, Bluetooth, and lights
2. Continuously monitor IR sensors
3. If entry sensor triggers first → increment count
4. If exit sensor triggers first → decrement count
5. Prevent negative count
6. Control lights based on count

7. Send updated data to Android app

Application and future scope:

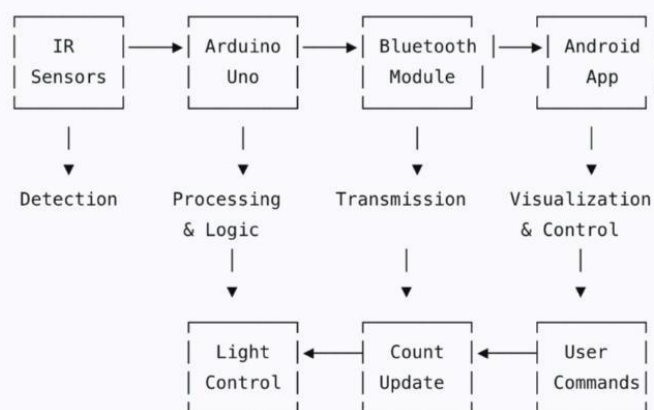
Applications

- Retail stores (customer counting)
- Offices and meeting rooms
- Libraries and classrooms
- Smart homes
- Hospitals and public buildings

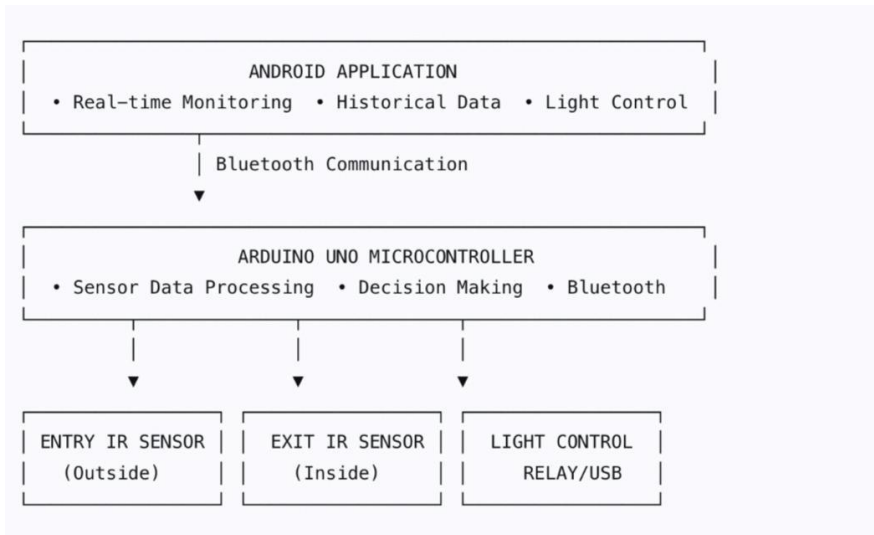
Future Scope

- Wi-Fi or cloud integration
- Multi-room or multi-zone counting
- Web dashboard and data analytics
- AI-based occupancy prediction
- Integration with access control systems
- Energy consumption optimization reports

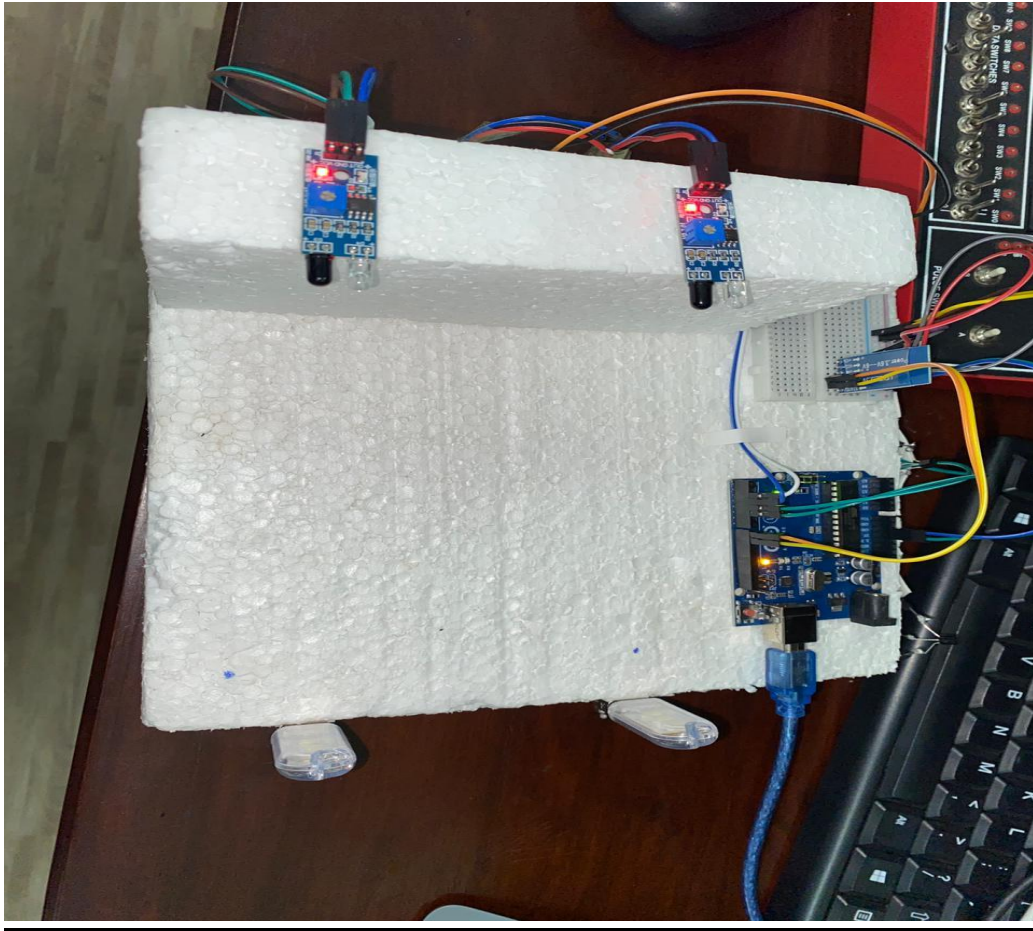
Data Flow Diagram:

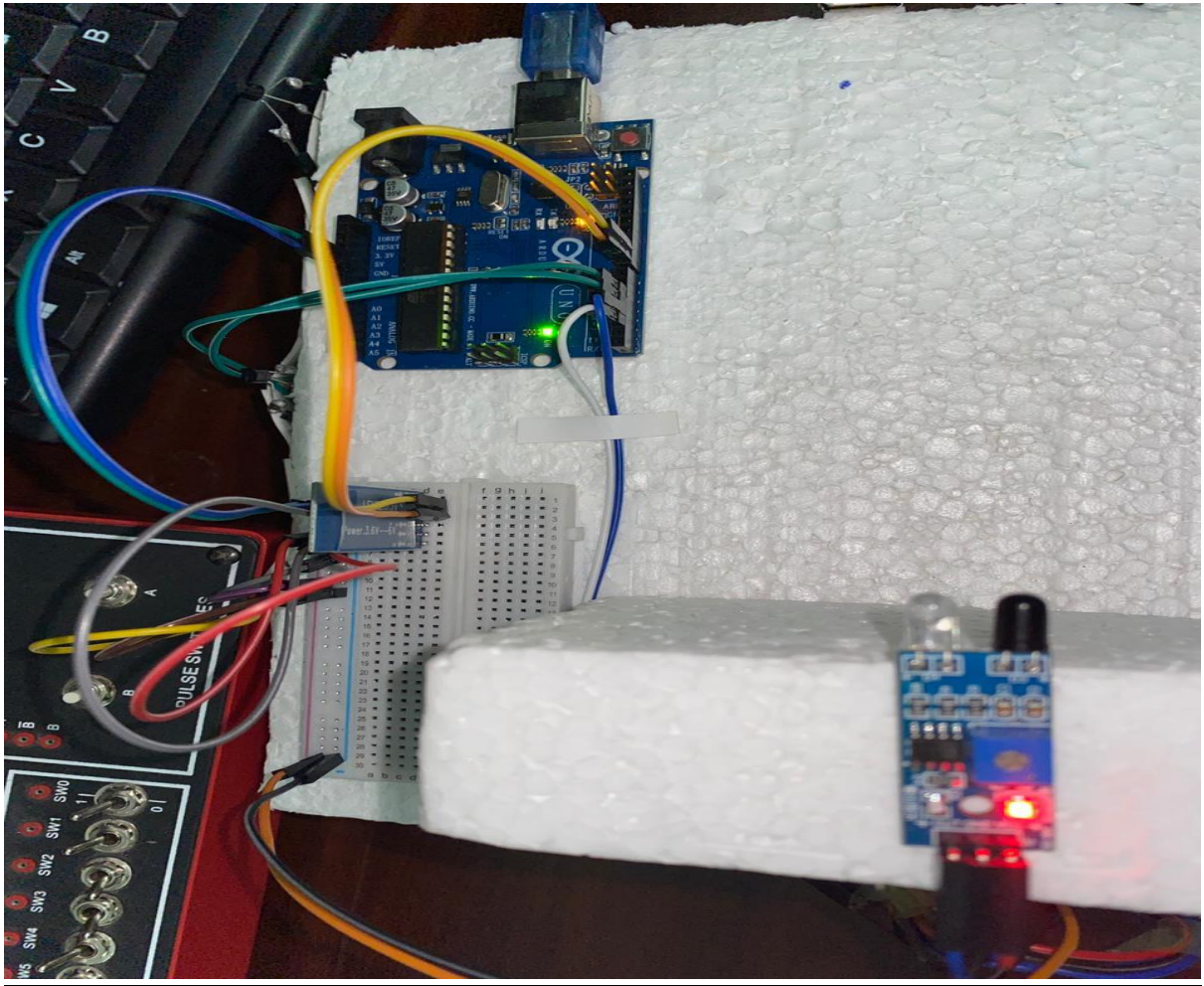


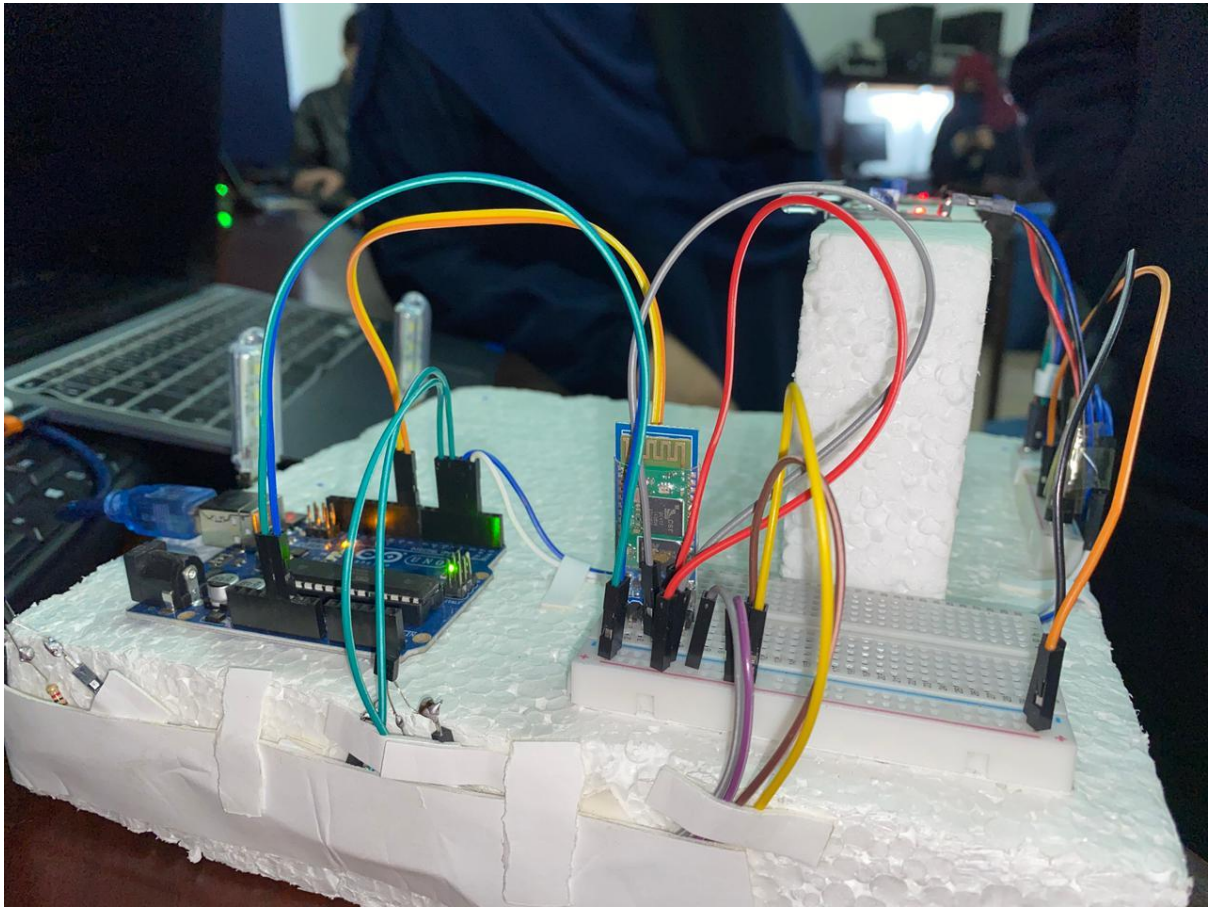
Block Diagram:



Practical Implementation:







Conclusion:

The **DLD Smart People Counter System with Automated Light Control** successfully demonstrates an efficient and low-cost solution for real-time occupancy monitoring and energy management. The system achieves high accuracy using a simple yet reliable algorithm and integrates hardware and software effectively.

This project not only fulfills its technical objectives but also provides a practical IoT-based solution that can be extended for larger smart-building applications. It serves as a strong educational and real-world implementation of Digital Logic Design and embedded systems concepts.

